

TIREE, United Kingdom

WMO: 031000

Lat: 56.4999N

Lon: 6.8807W

Elev: 12

StdP: 101.19

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.0	1.1	-4.8	2.5	2.0	-3.7	2.8	2.6	21.1	6.4	19.5	6.5	4.2	60	0.827

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	18.5	15.6	17.2	14.9	16.4	14.3	16.2	17.7	15.5	16.7	14.9	15.9	5.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.0	16.8	14.9	10.6	16.1	14.3	10.2	15.5	45.4	17.9	43.3	16.7	41.7	15.9	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.0	15.6	14.1	DB	-2.1	21.6	1.3	2.0	-3.0	23.0	-3.8	24.2	-4.5	25.3	-5.5	26.8	(5)
(5)			WB	-3.1	17.7	1.3	1.2	-4.1	18.6	-4.8	19.3	-5.6	19.9	-6.5	20.8	(6)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	6.0	5.8	6.4	8.0	10.0	12.1	13.6	13.9	12.8	10.6	8.2	6.5
	DBStd	3.54	2.06	2.15	2.03	2.01	2.07	1.64	1.52	1.45	1.57	2.05	2.27	2.50
	HDD10.0	638	126	119	112	66	25	2	0	0	1	18	62	109
	HDD18.3	3228	384	352	370	312	259	188	146	139	167	241	305	366
	CDD10.0	452	0	0	0	4	24	64	112	119	84	36	7	1
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.2	8.9	8.6	7.7	7.0	6.2	5.9	5.3	5.7	6.7	7.8	8.1	8.5
Precipitation	PrecAvg	1204	130	93	91	67	62	65	81	98	114	139	130	132
	PrecMax	1521	259	246	181	155	174	136	193	230	227	250	240	269
	PrecMin	856	17	6	5	8	15	23	20	11	27	37	44	41
	PrecStd	161	51	50	39	32	31	24	35	46	45	53	43	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.0	11.6	15.1	19.4	19.6	20.7	20.8	17.5	15.4	13.6	12.2
		MCWB	10.2	10.1	9.7	10.8	14.1	15.8	17.2	17.2	14.9	13.6	12.5	11.3
	2%	DB	10.3	10.3	10.5	12.7	16.0	17.1	18.2	18.1	16.4	14.6	12.7	11.5
		MCWB	9.6	9.5	9.2	10.0	12.7	14.6	15.7	15.8	14.4	13.2	11.7	10.7
	5%	DB	9.8	9.7	9.9	11.6	14.2	15.7	17.0	17.0	15.7	14.1	12.1	10.8
		MCWB	9.1	8.9	8.8	9.5	11.6	13.5	14.9	15.0	14.0	12.8	11.2	10.1
	10%	DB	9.2	9.1	9.4	10.8	13.0	14.7	16.1	16.2	15.2	13.5	11.5	10.2
		MCWB	8.4	8.2	8.3	9.1	10.9	12.9	14.3	14.3	13.7	12.3	10.4	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.5	10.4	11.6	14.7	16.2	17.4	17.5	15.7	14.4	13.1	11.6
		MCDB	10.6	10.7	10.9	13.6	18.1	18.9	19.9	19.9	16.4	14.9	13.4	11.9
	2%	WB	9.9	9.7	9.7	10.6	13.2	14.8	16.2	16.3	15.2	13.7	12.1	10.9
		MCDB	10.2	10.1	10.1	12.0	15.4	16.6	17.7	17.7	15.8	14.2	12.4	11.3
	5%	WB	9.3	9.1	9.1	10.0	12.1	13.9	15.3	15.5	14.6	13.2	11.4	10.3
		MCDB	9.7	9.6	9.7	11.1	13.6	15.2	16.5	16.6	15.4	13.8	11.9	10.8
	10%	WB	8.5	8.4	8.6	9.5	11.3	13.2	14.7	14.9	14.1	12.6	10.6	9.5
		MCDB	9.1	9.0	9.2	10.5	12.6	14.4	15.7	15.8	14.9	13.4	11.3	10.2
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.7	4.2	4.7	5.2	4.5	4.6	4.6	4.4	4.0	3.8	3.8
		MCDBR	3.7	3.4	4.0	5.4	6.7	5.8	5.6	5.4	4.6	3.7	3.6	3.8
		MCWBR	4.1	3.5	3.5	3.8	4.4	3.9	3.7	3.7	3.5	3.4	4.0	4.2
	5% WB	MCDBR	3.7	3.3	3.8	4.5	5.8	5.3	5.1	4.8	4.0	3.5	3.6	3.8
		MCWBR	4.2	3.6	3.7	3.5	4.3	3.7	3.7	3.7	3.5	3.8	4.3	4.4
Clear-Sky Solar Irradiance	taub	0.353	0.359	0.375	0.385	0.389	0.378	0.381	0.369	0.370	0.353	0.339	0.335	
	taud	2.154	2.259	2.298	2.327	2.372	2.446	2.447	2.487	2.456	2.386	2.233	2.135	
	Ebn at Noon	496	681	777	830	851	865	854	842	783	687	517	410	
	Edh at Noon	60	80	97	109	111	104	102	92	83	69	55	48	
All-Sky Solar Radiation	RadAvg	0.42	1.08	2.29	3.96	5.14	5.26	4.85	3.92	2.59	1.36	0.55	0.28	
	RadStd	0.04	0.10	0.16	0.26	0.38	0.53	0.33	0.26	0.16	0.14	0.05	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page